

Software-Centric SERVICES Transforming CLOUD COMPUTING

DEEPSHIKHA SHUKLA

SaaS refers to business software applications delivered via the cloud. As industries embrace new technologies and innovate to improve operations and solve customer problems, the move to a software-centric model will drive digital transformation

Software as a Service (SaaS) is a software service model in which the service provider hosts applications over the Internet to make these available to customers. It is the topmost layer of the cloud computing system, which has Platform as a Service (PaaS) and Infrastructure as a Service (IaaS) as the other two layers.

SaaS eliminates the expense of provisioning and maintenance, hardware acquisition, software licensing, installation and support for organisations. It also removes the need to install and run applications on their own computers or data centres.

SaaS is closely related to software delivery models like on-demand computing and application service providers. It allows users to use cloud-based apps such as email, calendar and office tools like Microsoft Office 365, Google apps, Salesforce, Citrix GoToMeeting and Cisco WebEx, among others.

SaaS enables device manufacturers to create value from software-enabled devices. This service-based licensing model can give a new lease of life to hardware applications all over the globe and facilitate flexible business models. SaaS can be used by enterprises for accounting and invoicing, tracking sales,

planning, performance monitoring, communications and more, in a number of industries.

How it works

The service provider hosts the customer's software and firmware over the Internet, by creating an application for SaaS distribution, and gives network-based access to registered customers. Whenever new features are implemented, these are updated for customers' access. During this process, customers' model data may be stored locally or in the cloud, as per the service level agreement (SLA).

SaaS applications can be integrated with other software using application programming interfaces (APIs). For example, a business can integrate SaaS provider APIs with its own written software tools.

Why go for the software-based approach

Before SaaS was available, companies had to purchase compact disks containing updates for software on their computers. For large organisations, updating software was a time-consuming process. Later, software updates became available for download on the Internet. This led companies to purchase additional licences and required a copy of the software to be installed on all devices.

SaaS is a Web-based model, in which the software vendor hosts and maintains servers, databases and codes that constitute an application. On-premise software is purchased through a perpetual licence, for which pay-per-year maintenance and support fees is needed. Whereas, SaaS has an annual or monthly subscription fee, which includes the software licence with support and most other fees.

SaaS is easy to implement, update and debug, and is less expensive. Users pay only for SaaS as-they-go, instead of purchasing multiple software licences for multiple computers. It has various uses

